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Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2028	8.27
Intermediate	GSHSEB	Citizen's School	2024	89.40%
Matriculation	GSHSEB	Citizen's School	2022	89.00%

Pursuing a Minor degree from the Centre for Machine Intelligence and Data Science (CMInDS), IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Ranked **42** of **5.7K+** globally (**11th** India) in WorldQuant BRAIN's Power Pool Alpha Competition Apr'25–May'25
- Secured **99.72** percentile in **Joint Entrance Examination-Mains** over **1.4 million** candidates nationwide '24
- Secured **98.36** percentile in **Joint Entrance Examination-Advanced** over **180,000+** candidates '24
- Secured admission to the prestigious **BS (Research)** Program at **IISc Bangalore**, among India's top applicants '24
- Ranked in the top **2.47%** over a pool of **48,000+** Group A candidates in the **Gujarat Common Entrance Test** '24

RESEARCH AND ENGINEERING EXPERIENCE

Self Driving Car [SeDriCa]

Sep'25 – Present

Team Unmesh Mashruwala Innovation Cell [UMIC] | Guide: Prof. Archak Mittal

IIT Bombay

Cross-functional team of **30+** students developing **India's first Level 5** self-driving car, customized for Indian roads.

- F1TENTH Autonomous Racing** | *Autonomous Controls* | *Developing a 1/10-scale Self-Driving Racecar*
 - Engineering **LibTorch-Acados C++** runtime to deploy learned-dynamics **HyperMPC** at **100 Hz** control loop
 - Developing a **GRU-based Hyper-Param** with **B-spline** control encoding to learn horizon-varying model parameters
- Autonomous Golf Cart** | *Localization & State Estimation* | *Building the Autonomy Stack for a Full-Scale Golf Cart*
 - Building an autonomous localization stack for reliable outdoor navigation using high-precision **RTK GNSS**
 - Developing **Joint EKF** fusing **GNSS & IMU** measurements to estimate vehicle pose, velocity & heading in real-time

BRAIN Research Consultant

Feb'25 – Dec'25

WorldQuant BRAIN | Quantitative Alpha Research

Remote

Selected as **1** of the **12 WQBRain Research Consultants** (Rising Stars cohort) to attend the **IQC'25 Finals**.

- Achieved **Master Consultant**(Q3'25) via portfolio of orthogonal signals yielding an after-cost out-sample **IR** of **2.39**
- Submitted **320+** alphas with out-sample fitness score upto **6.47**, maintaining average **self-correlation** of **0.25**
- Engineered **90+** **composite alpha strategies** by selecting, filtering & combining orthogonal quantitative signals
- Automated alpha research via **LLM-driven** iterative refinement pipeline for large-scale quantitative signal generation

KEY PROJECTS

Regime Aware Portfolio Optimization

| Summer Of Quant, Quant Club

Jun'26 – Jul'26

- Built a regime-switching portfolio optimizer integrating **HMMs**, **Random Forest** meta-models & convex allocation
- Built event-driven trading labels using **CUSUM** sampling, **Triple Barrier** labeling & volatility-normalized thresholds
- Constructed **250+** alpha features spanning price, volatility, momentum, volume & macroeconomic factor spaces
- Mitigated temporal leakage across **503** equities via **Purged 5-Fold CV**, **1%** embargoes & sequential bootstrapping

The Cosmic Detective

| Krittika Summer Projects, Krittika Astronomy Club

May'26 – Present

- Created **multi-agent AI agents** in **7-member** team for **autonomous sky exploration & hypothesis generation**
- Built an astronomy benchmark pipeline integrating **Gemini** with literature retrieval & fault-tolerant LLM orchestration
- Constructed a semantic **Neo4j** knowledge graph using **transformer-based** claim de-duplication with **95%** similarity
- Engineered a **Swift FITS** pipeline extracting temporal, spectral & physical features from astronomical observations

Single Image Super Resolution

| ME 228, Course Project: Neeraj Kumbhakarna

Mar'26 – Apr'26

- Built an **EDSR** with **32 residual blocks**, boosting training via mixed-precision (FP16/FP32) using **PyTorch AMP**
- Trained the model over **290k iterations** using **L1 loss**, achieving a **34.84 dB PSNR** & **0.939 SSIM** on **DIV2K**
- Bounded peak VRAM spikes via **spatial tiling** & **AMP**, deploying a **sub-pixel convolution** for fast upsampling
- Benchmarked an **SRCNN** on **Sen2Venus** data, securing **37.47 dB PSNR** to map non-linear **LR-to-HR** features

Inverse Elasticity Physics Informed NN

| ME218, Course Project: Sripriya Ramamoorthy

Mar'26 – Apr'26

- Developed a **Direct Hybrid Collocation PINN** estimating **Young's modulus** fields across **3 AM** lattice specimens
- Unified **8 DIC anchors** with **10,000+** **Sobol** collocation points, enforcing **2D Cauchy Momentum** PDE constraints
- Solved inverse elasticity problem using a **7-objective PINN** to reconstruct displacement, stress, strain & modulus fields

Crypto Strategy Optimizer

| Seasons of Code, Web and Coding Club

May'25 – Jul'25

- Engineered a mid-frequency **Bitcoin options** simulator with **slippage-aware** logic & **Delta Exchange** integration
- Built an event-driven execution engine for asynchronous **ccxt API** interactions, minimizing execution overhead
- Designed dynamic strike selection & trade lifecycle management with automated **PnL**, turnover, slippage & analytics
- Deployed **Multi-Armed Bandits** for hyperparameter tuning, balancing exploration-exploitation for fast convergence

- Engineered feature vectors (**HOG, GLCM, LAB**) to isolate texture variance, distinguishing signal from background
- Ranked features via **Random Forest**, reducing **750+ features** and built **Active Learning** pipeline with **SMOTE**
- Optimized **XGBoost** hyperparameters via **GridSearchCV** & achieved **0.85 F1 Score** on highly skewed datasets

OTHER PROJECTS

Effects of Interest Rate Hikes on Financial Markets | *FINSEARCH, Finance Club* Jun'26 – Jul'26

- Analyzed **RBI's** past years' interest rate hikes & their macroeconomic effects on **economic growth & inflation**
- Compiled **20+ years** of RBI monetary policy data, identifying major tightening cycles & their macroeconomic drivers
- Applied the **Vasicek interest rate model** to simulate & assess **bond price sensitivity** during RBI rate hike cycles

Spatial Hyper-MPC with Neural Net | *ME444, Course Project: Seshu Pasumarthy* Mar'26 – Apr'26

- Developed a hybrid **Neural-MPC** controller for a **6-DOF suspension** using **OSQP, Joint EKF** & residual learning
- Distilled a **40-step** MPC into a **4,736-parameter** neural policy using **20K** expert trajectories for real-time inference
- Benchmarked against tuned **MIMO PID** across **3** simulation scenarios, reducing peak angular velocities by up to **3×**

Spatial Active Noise Control | *ME791, Course Project: Sripriya Ramamoorthy* Mar'26 – May'26

- Built **virtual sensing** framework for **Spatial ANC** using signals from **4** remote microphones for pressure reconstruction
- Estimated virtual microphone pressure by training an **ObsTasNet** observation-filter on **400** simulated acoustic scenes
- Benchmarked against analytical kernel interpolation, achieving **12.6×** faster virtual microphone pressure estimation

Mathematics of Derivative Pricing | *Summer Of Science, Maths & Physics Club* May'25–Jul'25

- Applied a Python framework for pricing **forwards, futures & options**, linking theory with implementation
- Implemented the **Black–Scholes model** for European options using stochastic processes & risk-neutral valuation
- Performed **sensitivity analysis** on **volatility, interest rates & maturity** to quantify impact on derivative pricing
- Explored advanced methods including **martingale pricing, hedging strategies & finite-difference techniques**

POSITION OF RESPONSIBILITIES

Team Manager | **Self Driving Car, Unmesh Mashruwala Innovation Cell** Apr'26 – Present

Cross-functional team of 30+ students developing India's first Level 5 self-driving car, customized for Indian roads.

- Responsible for effectively managing an annual budget of **INR 4+ Million**, allocating resources for optimal outcomes
- Coordinating communication between the Technical Core & Non-Technical teams to ensure effective collaboration
- Spearheaded a **2-month** recruitment drive for **100+** aspirants, including interviews, training & project evaluation
- Mentoring technical & non-technical teams while preparing for upcoming national & international competitions

Student Mentor | *Summer Of Science, Maths and Physics Club* May'26–Jul'26

- Guided **9** students with suitable references & materials to build theoretical understanding of **Algorithmic Trading**

Project Mentor | Build the Algorithmic Black Box, *Winter in Data Science, Analytics Club* Dec'25–Jan'25

- Mentored **20+** students in building **algo-trading protocols** & validating **system dynamics** within trading engines

TECHNICAL SKILLS

Programming	Python, C++ (STL), C, MATLAB, Git, Docker, Linux, SQL, Django, HTML, CSS, LaTeX
Libraries	Numpy, Matplotlib, Pandas, Scikit-Learn, TensorFlow, PyTorch, CUDA, NLTK, Keras
Software Packages	ROS2, MS Excel, Cloud Computing, Autodesk Fusion, Gazebo, Arduino IDE, EasyEDA

COURSES UNDERTAKEN

Data Science & Programming	Statistical Machine Learning and Data Mining†, Applied Data Science & Machine Learning, Programming in Data Science, Computer Programming & Utilization
Mathematics	Calculus, Linear Algebra & Differential Equations
Engineering	Numerical Analysis, Control Systems, Analysis and Design of Mechanical Systems, Acoustic Devices
Miscellaneous	Social Learning & Herding, Introduction to Financial Engineering, Economics, Biology, Makerspace, Design Thinking for Innovation

† to be completed by Nov'26

EXTRACURRICULAR ACTIVITIES

- Built a **goal-based** multi-asset portfolio using quantitative equity screening & liability investing in **PMC'26** Jun'26
- Learned **Big Data Handling with EXCEL & SQL** (Analytics Club), through the Learners' Space Jun'26–Jul'26
- Built **LexAI**, Securing a **Top-10 finish** among competing cohorts at the **AIC X Anthropic** Claude Hackathon May'26
- Mentored **12** participants across **3** teams for **CodeWars V6**, WnCC event, leading one team to be the **finalist** Mar'26
- Completed the **DSA Bootcamp** organised by the **WnCC** in collaboration with **GeeksForGeeks** Jun'25–Jul'25
- Learned **Data Science Fundamentals** (Analytics Club), **Natural Language Processing** (WnCC Club), **Hand Written Digit Recognition Using CNN** (AIC), through the courses offered in **Learners' Space 2025** Jul'25
- Developed **valuation models**, covering **corporate finance & analysis** in Financial Modeling bootcamp Dec'24–Jan'25
- Built a **drone controller**, covering **design & circuitry integration** in Makerspace (MS101) project Aug'24–Nov'24
- Led a team of 4 to build a **gesture-controlled WiFi bot** for **XLR8** using RPi Pico W, ESP01 & L298N Sep'24
- Contributed to **sustainability & community** initiatives with NSS's **ENS** department for about **8 months** '24-'25
- Participated in the **EnB Buzz** competition by E-Cell, pitching a start-up idea & presenting a business model Oct'24